WORKED SOLUTION

STEP II 2018, Question 5

Pure • Integration of Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q5

The key idea

Each part turns a function into a power series and integrates it one term at a time. In (i) that is just integrating a geometric series. In (ii), after expanding the exponential, every term becomes an integral of the form $\int_0^\infty x^m \mathrm{e}^{-x}\,\mathrm{d}x$, which is a factorial, and the factorials cancel to leave exactly the series from (i). For (iii), the $\ln x$ in the denominator is the clue: substituting $x = \mathrm{e}^{-t}$ turns it into $-t$ and turns the integral into two copies of the integral in (ii).

Solution

Part (i)

For $|x| < 1$ the binomial expansion (a geometric series with ratio $-x$) is

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots = \sum_{n=0}^{\infty} (-x)^n = \sum_{n=0}^{\infty} (-1)^n x^n.$$

Integrate both sides from 0 to x. On the left, $\int_0^x \frac{\mathrm{d}u}{1+u} = \ln(1+x)$, since $\ln 1 = 0$. On the right, integrating term by term,

$$\begin{aligned} \ln(1+x) &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} \qquad \text{(replacing } n+1 \text{ by } n\text{)} \\ &= -\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n} = -\sum_{n=1}^{\infty} \frac{(-x)^n}{n}, \end{aligned}$$

which is the required result for $|x| < 1$.

Part (ii)

For all x,

$$\mathrm{e}^{-ax} = \sum_{n=0}^{\infty} \frac{(-ax)^n}{n!} = 1 - ax + \frac{a^2x^2}{2!} - \frac{a^3x^3}{3!} + \cdots.$$

The $n = 0$ term is 1, so

$$\frac{1 - \mathrm{e}^{-ax}}{x} = -\sum_{n=1}^{\infty} \frac{(-a)^n x^{n-1}}{n!}.$$

Dividing by x is harmless here: every remaining term has a factor x, so the integrand is perfectly well behaved at $x = 0$ (it tends to a). Multiplying by e^{-x} and integrating term by term (we are told to ignore convergence),

$$\int_0^{\infty} \frac{(1 - \mathrm{e}^{-ax})\,\mathrm{e}^{-x}}{x}\,\mathrm{d}x = -\sum_{n=1}^{\infty} \frac{(-a)^n}{n!} I_{n-1}, \qquad \text{where } I_m = \int_0^{\infty} x^m \mathrm{e}^{-x}\,\mathrm{d}x.$$

Ask yourself What is I_m?

Integrate by parts with $u = x^m$ and $\mathrm{d}v = \mathrm{e}^{-x}\,\mathrm{d}x$. For $m \geq 1$,

$$I_m = \Big[-x^m \mathrm{e}^{-x} \Big]_0^{\infty} + m \int_0^{\infty} x^{m-1} \mathrm{e}^{-x}\,\mathrm{d}x = m I_{m-1},$$

because $x^m \mathrm{e}^{-x} \to 0$ as $x \to \infty$ (the exponential wins) and is 0 at $x = 0$. Since $I_0 = \int_0^{\infty} \mathrm{e}^{-x}\,\mathrm{d}x = 1$, it follows that $I_m = m!$.

So $I_{n-1} = (n-1)!$, and $\dfrac{(n-1)!}{n!} = \dfrac{1}{n}$. Therefore

$$\int_0^{\infty} \frac{(1 - \mathrm{e}^{-ax})\,\mathrm{e}^{-x}}{x}\,\mathrm{d}x = -\sum_{n=1}^{\infty} \frac{(-a)^n}{n} = \ln(1 + a),$$

using part (i) with $x = a$, which is allowed because $|a| < 1$. This is the required result.

Part (iii)

Substitute $x = \mathrm{e}^{-t}$, so that $\ln x = -t$ and $\mathrm{d}x = -\mathrm{e}^{-t}\,\mathrm{d}t$. As x runs from 0 to 1, t runs from ∞ down to 0. Also $x^p = \mathrm{e}^{-pt}$ and $x^q = \mathrm{e}^{-qt}$. Hence

$$\int_0^1 \frac{x^p - x^q}{\ln x}\,\mathrm{d}x = \int_{\infty}^0 \frac{\mathrm{e}^{-pt} - \mathrm{e}^{-qt}}{-t}\left(-\mathrm{e}^{-t}\right)\mathrm{d}t = \int_0^{\infty} \frac{\left(\mathrm{e}^{-qt} - \mathrm{e}^{-pt}\right)\mathrm{e}^{-t}}{t}\,\mathrm{d}t,$$

where the two minus signs cancelled and swapping the limits produced the swap of p and q. Now write $\mathrm{e}^{-qt} - \mathrm{e}^{-pt} = \left(1 - \mathrm{e}^{-pt}\right) - \left(1 - \mathrm{e}^{-qt}\right)$ and split the integral. Each piece has the form in (ii), with $a = p$ and $a = q$ respectively, and both satisfy $|a| < 1$. So

$$\int_0^1 \frac{x^p - x^q}{\ln x}\,\mathrm{d}x = \ln(1 + p) - \ln(1 + q) = \ln\left(\frac{1 + p}{1 + q}\right).$$

Splitting is legitimate because each of the two integrals converges on its own: by (ii) each has a finite value.

Checks

- If $p = q$ the integrand is zero, and indeed $\ln \frac{1+p}{1+p} = 0$.

- Swapping p and q changes the sign of both the integrand and the answer.
- Sign check: for $0 < x < 1$, $\ln x < 0$; if $p < q$ then $x^p > x^q$, so the integrand is negative, and so is $\ln \frac{1+p}{1+q}$.
- With $p = \frac{1}{2}$, $q = 0$ the answer is $\ln \frac{3}{2} \approx 0.405465$. A numerical integration of $\int_0^1 \frac{\sqrt{x}-1}{\ln x}\,\mathrm{d}x$ gives the same value, and random choices of p and q in $(-1,1)$ agree with the formula to six decimal places.

Common slips

- Forgetting that the $n = 0$ term of e^{-ax} cancels the 1, and then trying to integrate e^{-x}/x, which diverges at 0.
- Losing a sign in the substitution in (iii): there are three minus signs to track (from $\ln x$, from $\mathrm{d}x$, and from reversing the limits), and getting one wrong gives $\ln \frac{1+q}{1+p}$.
- Writing $I_{n-1} = n!$ instead of $(n-1)!$; then the factorials no longer cancel to $\frac{1}{n}$.

Beyond STEP

The integral $I_m = \int_0^\infty x^m \mathrm{e}^{-x}\,\mathrm{d}x = m!$ is the gamma function $\Gamma(m+1)$, which extends the factorial to non-integers. The integral in (iii) is a classic of "differentiating under the integral sign": differentiating with respect to p removes the awkward $\ln x$ and leaves $\int_0^1 x^p\,\mathrm{d}x = \frac{1}{p+1}$. The result actually holds whenever $p, q > -1$. Swapping a sum and an integral, which we were told not to worry about, is justified in analysis courses by uniform or dominated convergence. For the analysis behind these steps, see Abbott, *Understanding Analysis*.

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